MILITARY SPECIFICATION

INDICATOR, AZIMUTH; ID-307/ARN

This amendment forms a part of Military Specification MIL-I-8691A(ASG) dated 17 October 1955 and has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

Page 1, paragraph 1.2, "Size" and "Weight" columns: Delete and substitute:

"Maximum Size (Inches)	Maximum Weight (Lbs)
3-1/4 square by 6-5/8 overall length	3.0 "

Pages 4 and 6, figures 1 and 2: Delete and substitute the attached figures 1 and 2, respectively.

Page 9, paragraph 3.4.9.2: Delete and substitute:

"3.4.9.2 Service (Environmental Conditions).- The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as otherwise specified herein."

Page 9, paragraph 3.4.9.2.2 (b), line 2: Delete "1.32" and substitute "3.4".

Page 10, paragraph 3.4.9.2.3 (b), line 2: Delete "1.32" and substitute "3.4".

Page 13, paragraph 4.2.1.2.2, line 4: Delete "±2" and substitute "±12".

Page 13, paragraph 4.2.1.2.4: Delete and substitute:

4.2.1.2.4 Bearing measuring potentiometer accuracy.- The resistance of the potentiometer card circuit as measured between pins k and m on a Wheatstone Bridge, shall measure 14,400 ohms ± 1.5 percent. With the pointer servoed to 0 ± 0.5 degrees by the direct servo drive to the harmonic resolver stator null, the sine-cosine potentiometer shall be set so that when the car is excited from a 30 cps ± 0.1 percent center tapped voltage, the phase shift between the voltage at pin K and the brush output at the junction of a 530,515 ohm ±1 percent resistor and a 0.01 microfarad ± 1 percent capacitor connected in series to the brushes of the potentiometer will be 62 ±1.5 degrees. The 530,515 ohm resistor shall be connected to pin P and the 0.01 microfarad capacitor connected to pin N with the center tap of the 30 cps supply voltage grounded in this circuit. The brush output junction to ground voltage shall lead the pin K to ground voltage by 62 degrees, with increasing lead angle resulting in increasing bearing indications of the indicator pointer. When tested as a phase shifter, the angular accuracy of the potentiometer shall be ± 1.5 degrees as the potentiometer is positioned by the direct servo drive in 30-degree bearing increments through one revolution of the indicator pointer starting from the 0 degree setting."

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Page 14, paragraph 4.2.1.2.5: Delete and substitute:

"4.2.1.2.5 Bearing measuring resolver accuracy.- With the pointer servoed to ±0.25 degrees by the harmonic resolver stator null, the bearing measuring resolv be so set that when the rotor is excited from a 270-ohm source impedance through 560 ohm ±1 percent resistor at a frequency of 135 ±1 cps, and when the magnitude voltage source is adjusted until the voltage between pin N and pin P is 0.7 volt phase shift will be 206 ±1 degree between the input into the 830-ohm resistor ar stator output with a 117,920 ohm ±1 percent resistor and a 0.01 microfarad ±1 pe capacitor connected in series. The 117,920-ohm resistor shall be connected to p and the 0.01 microfarad capacitor to pin R. Pins P and S shall be grounded in t circuit. The common resistor and capacitor junction output shall lead the input into the 830-ohm resistor by 206 ±1 degree with decreasing lead angle resulting decreasing pointer indications. The null between pin R and pin S shall occur at approximately 351 degrees. The error in angular accuracy measured for every 5 c of the indicator pointer for one revolution of the resolver and every 30 degrees indicator pointer for one revolution of the pointer and determined by the resol. shall not exceed a spread of 1.6 degrees when tested by feeding the harmonic re. output to a servo amplifier which will drive the indicator to the null position harmonic resolver."

Custodians:
Navy - Bureau of Aeronautics
Air Force

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